Project Proposal

Project Title: *Animexia AI: Design and Development of an Intelligent Anime and Manga Recommendation System*

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Project Overview

Animexia AI is an intelligent, domain-specific Artificial Intelligence system designed to serve as a personalized anime and manga companion. The system leverages curated datasets from platforms such as MyAnimeList (MAL) and Kaggle to provide users with accurate information, recommendations, ratings analysis, and interactive discussions related to anime and manga content. Unlike generic chatbots, Animexia AI focuses on factual accuracy, structured knowledge retrieval, and user-centric recommendation logic, making it a specialized conversational AI for anime enthusiasts.

Motivation and Significance

Anime and manga fans often rely on multiple platforms to gather information, ratings, reviews, and recommendations. These sources are fragmented and may lack personalization or conversational interaction. Existing AI systems are mostly general-purpose and do not offer domain-specific depth or factual grounding.

Animexia AI addresses this gap by combining structured datasets, recommendation logic, and natural language interaction to deliver a unified, reliable, and engaging experience. The project demonstrates the practical application of AI concepts such as data preprocessing, information retrieval, recommendation systems, and conversational interfaces.

Scope and Applications

The scope of Animexia AI includes anime and manga recommendation, factual information retrieval, rating comparison, and interactive discussion. The system can be used by anime fans, beginners seeking recommendations, content reviewers, and for educational demonstrations of applied AI. It also serves as a foundational model for future domain-specific conversational agents.

System Architecture

Animexia AI operates through a structured AI pipeline consisting of data collection, preprocessing, knowledge representation, and user interaction. Datasets sourced from Kaggle and MyAnimeList provide anime and manga metadata such as genres, ratings, popularity, studios, and descriptions.

User queries are processed using natural language techniques and mapped to relevant dataset attributes. Recommendation logic filters and ranks anime based on user preferences, popularity scores, and factual ratings. The system responds with informative, context-aware outputs while maintaining a conversational tone. This architecture ensures both accuracy and engagement.

Datasets and Tools Used

- **Primary Database:** MyAnimeList (MAL) – ratings, popularity, genres, reviews.
- **Secondary Datasets:** Kaggle anime and manga datasets (metadata, user ratings).
- **Programming Language:** Python.
- **Libraries and Frameworks:**
  - *Pandas, NumPy* – Data processing and manipulation.
  - *Scikit-learn* – Recommendation and similarity models (Cosine Similarity, KNN).
  - *NLP Libraries* – Query understanding and tokenization.
- **Development Environment:** Google Colab / Jupyter Notebook.

Intelligence Components

User preferences, ratings, and interaction history act as feedback inputs to refine recommendations and improve response relevance over time, creating a rudimentary feedback loop for continuous improvement.

Conclusion

Animexia AI presents a creative yet academically grounded application of Artificial Intelligence focused on a popular real-world domain. By integrating structured datasets, recommendation algorithms, and conversational interaction, the project showcases how AI can be tailored to specific user communities. The system highlights the effectiveness of domain-specific intelligence over generic models, making it both practical and engaging.

Future Enhancements

To ensure the longevity and scalability of the project, the following enhancements are proposed:

- Integration of a fine-tuned Large Language Model (LLM) for richer conversations.
- User profile-based personalization and long-term preference learning.
- Sentiment analysis on reviews and discussions.
- Web or mobile-based user interface (Streamlit/React).
- Expansion to include seasonal anime trends and release predictions.